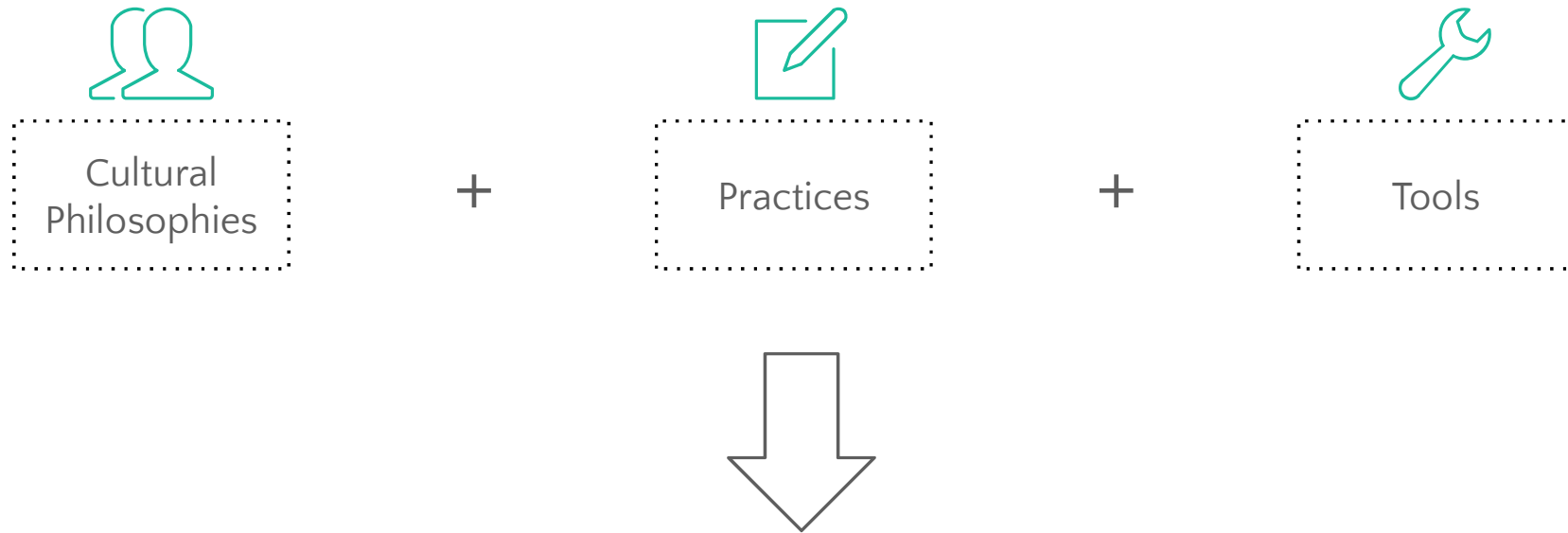




Introduction to DevOps

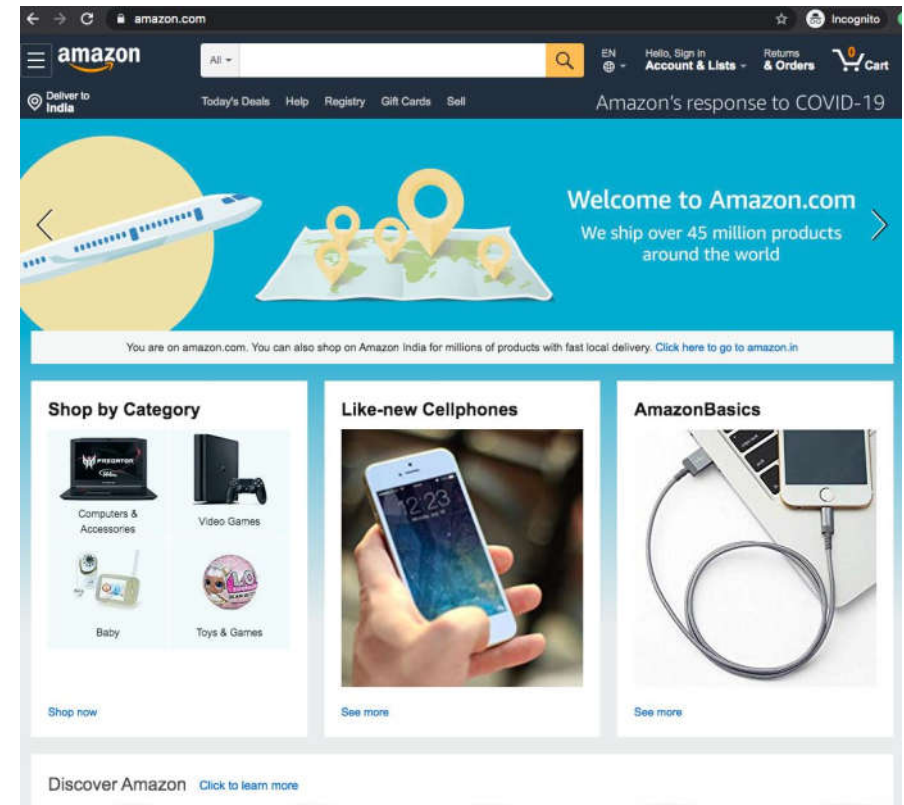
What is DevOps



To Deliver applications and services at high velocity

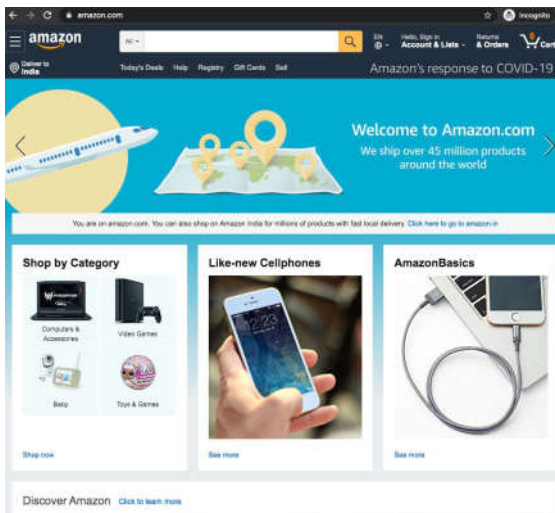
Why DevOps? – Scenario

How will we build a website like Amazon.com?



Why DevOps? – Scenario

How will we build a website like Amazon.com to serve lots of users?



- Serving huge number of users
- With so many complex features
- Delivering the bug fixes quickly

Why DevOps? – Basic Solution

How will we build a website like Amazon to serve lots of users?

Approach 1:

- Understand the requirements
- Write the perfect code once – software
- Deploy it on very powerful server
- Fire the software development team! 😊😊

Why DevOps? – Basic Solution

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What challenges do you see with this approach?

Did we miss anything?

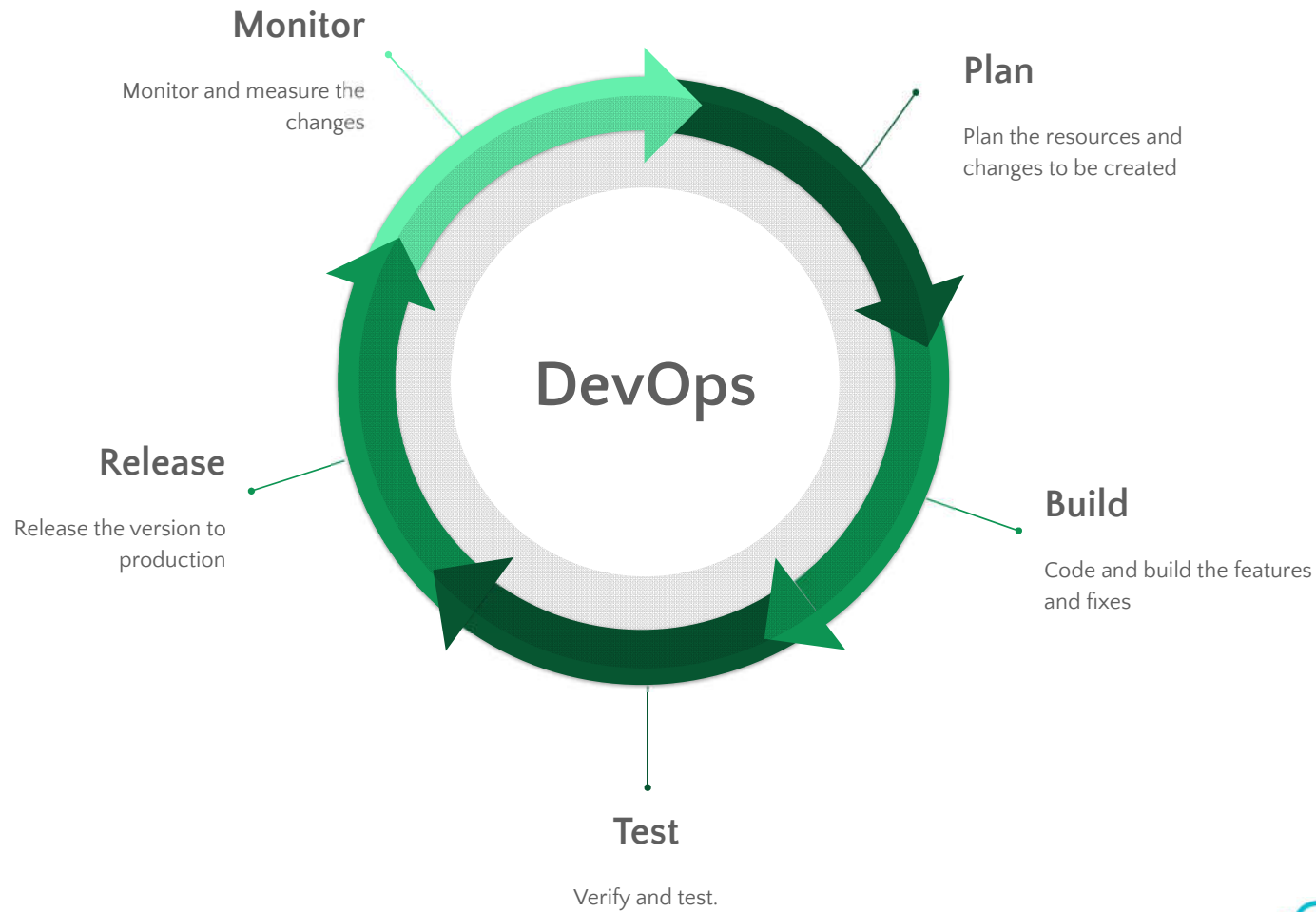
Why DevOps? – Basic Solution

How will we build a website like Amazon to serve lots of users?

Challenges with first Approach:

- Software development is a continuous life long process not one time
- We must build the software in incremental phases.
- Collaboration between many software engineers is involved
- Testing needs to be the first thing not last
- Deployment needs to be done with more thorough approach to multiple machines

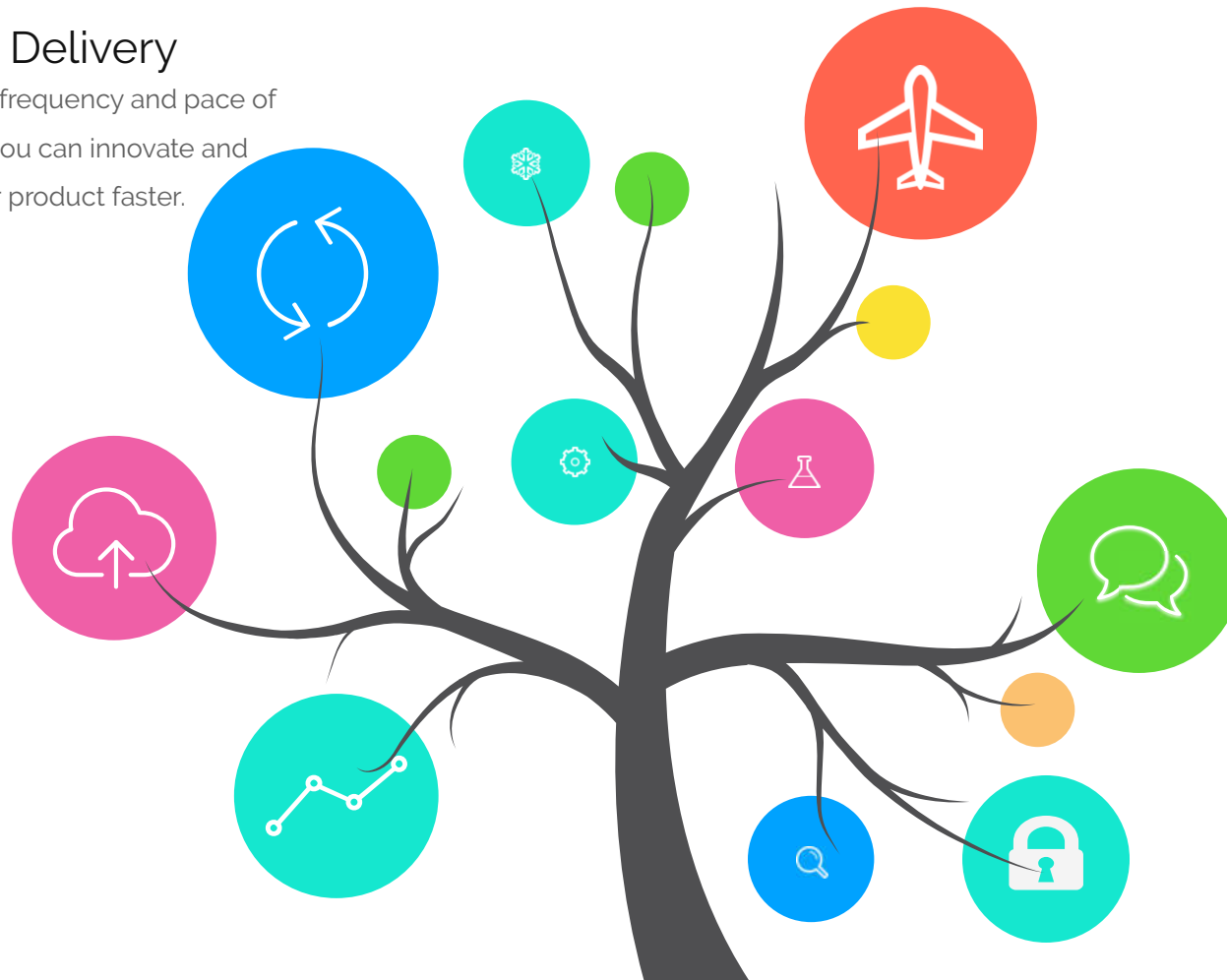
DevOps – Solution



Benefits of Devops

1. Rapid Delivery

Increase the frequency and pace of releases so you can innovate and improve your product faster.



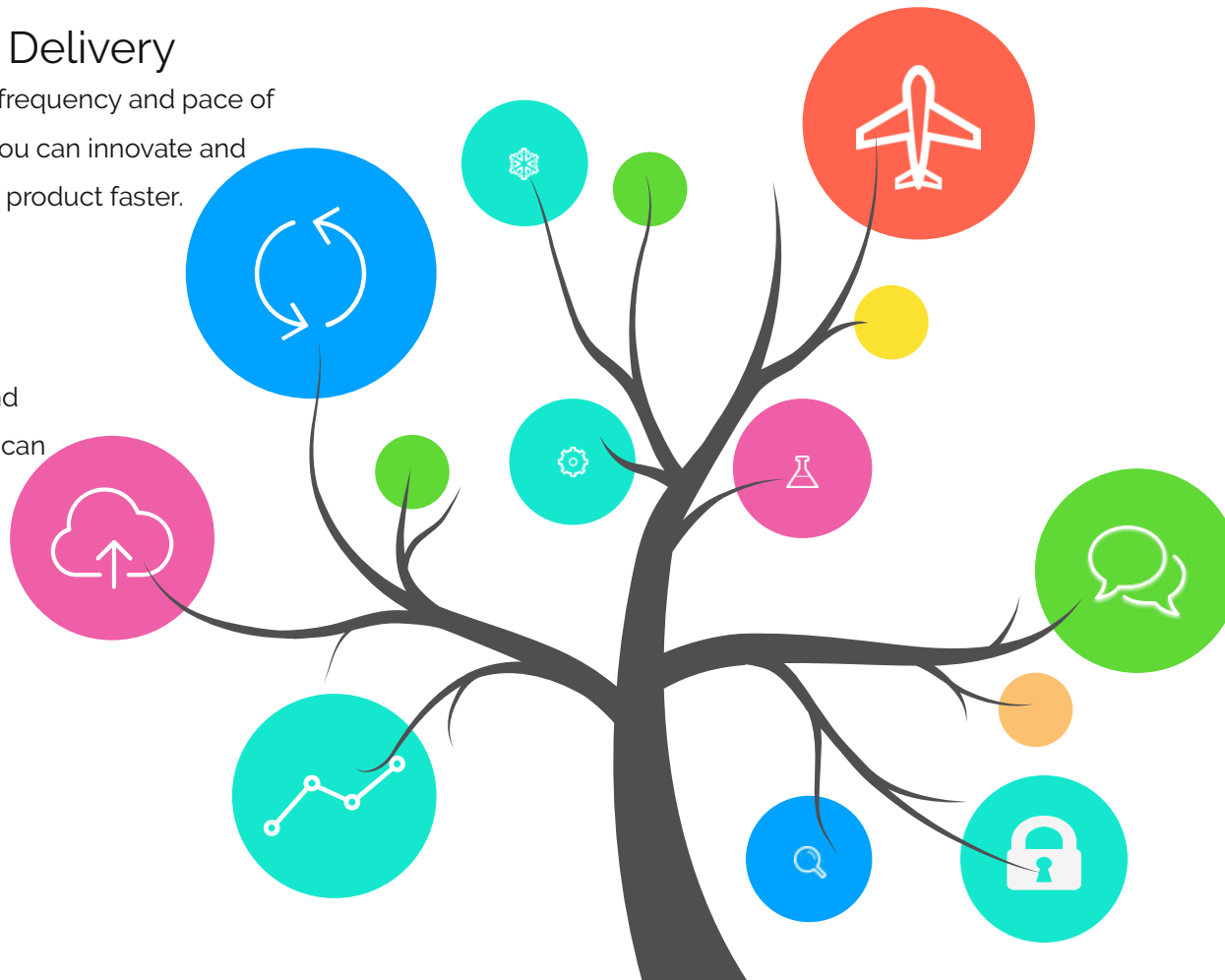
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Benefits of Devops

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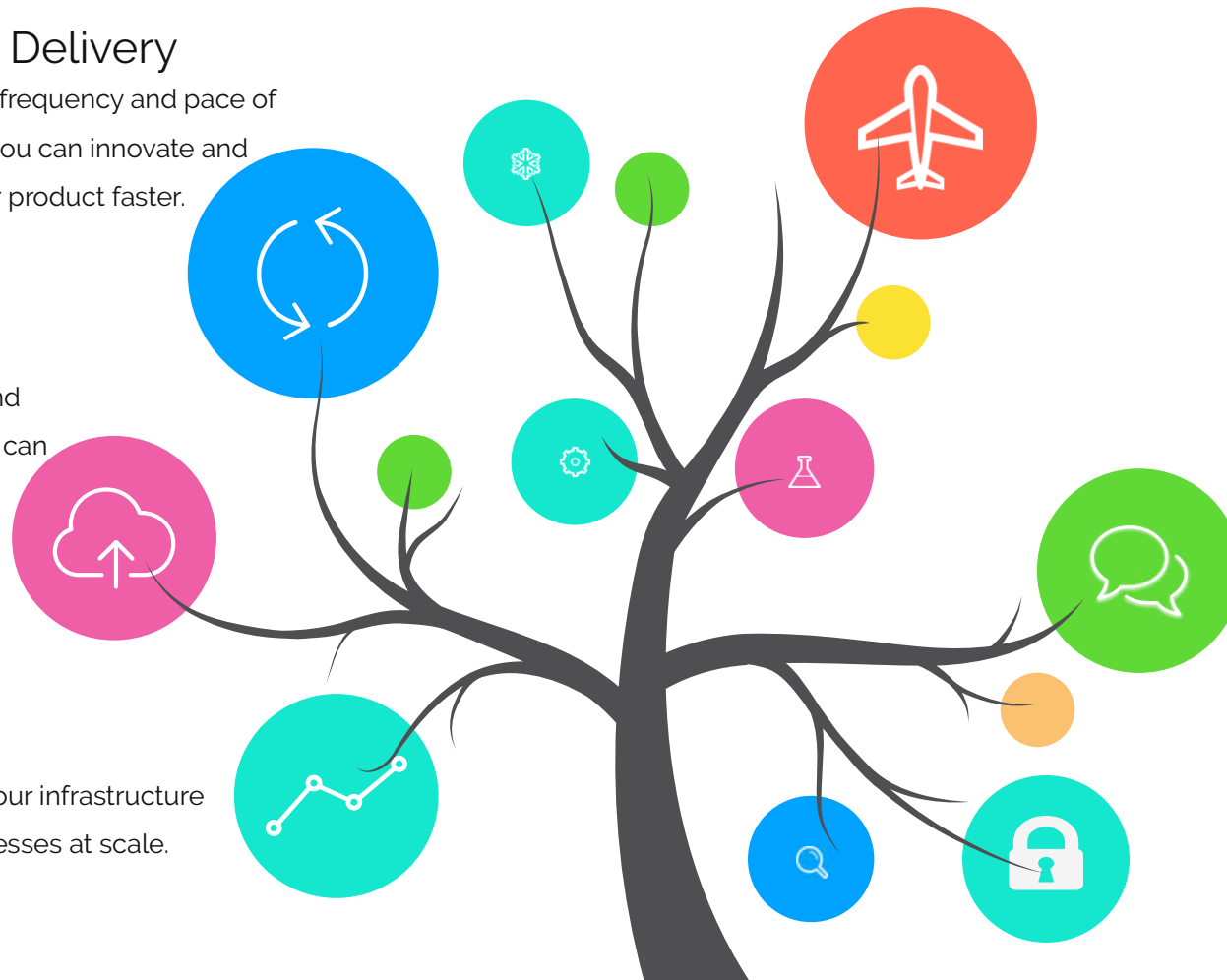
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Operate and manage your infrastructure and development processes at scale.



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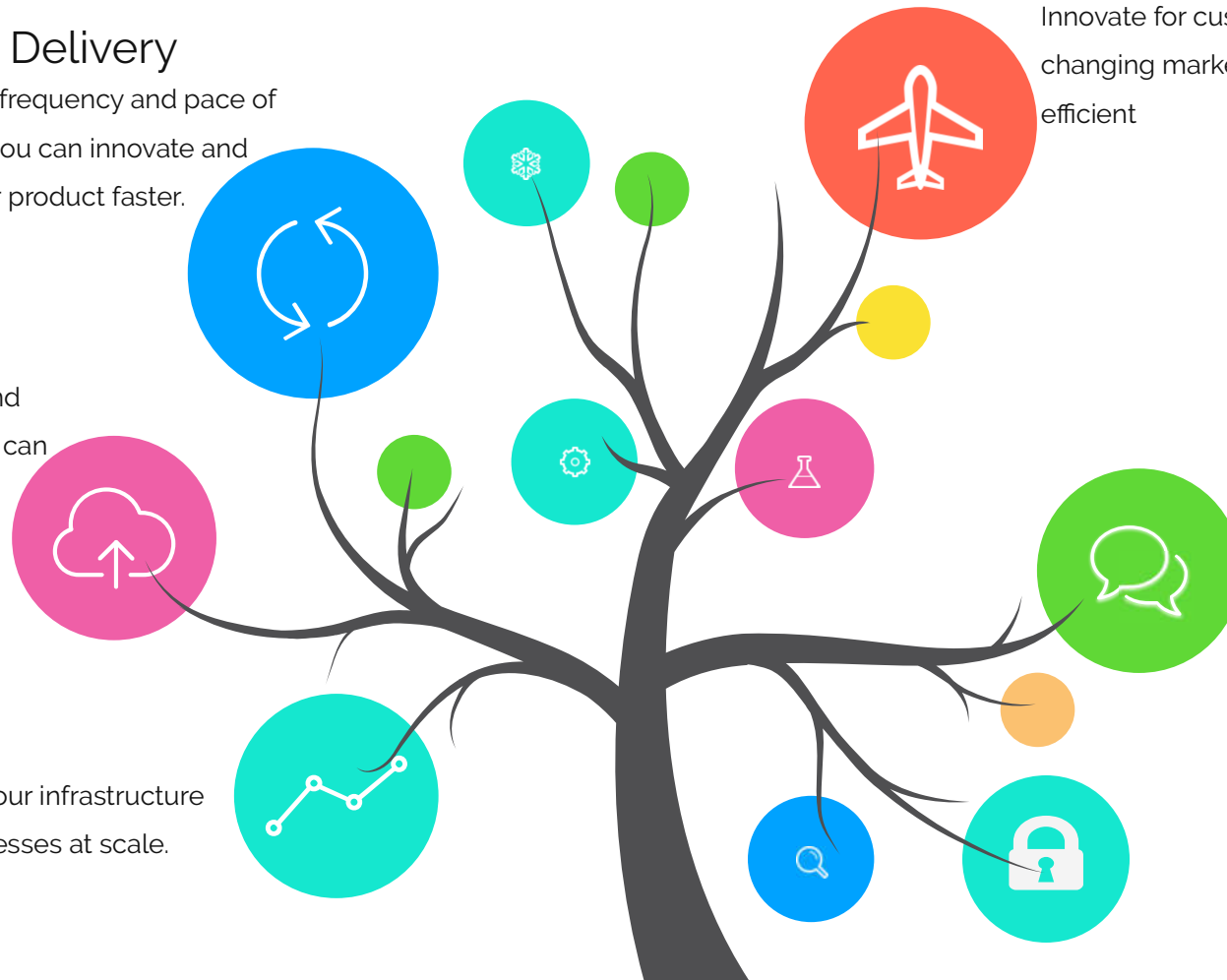
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Innovate for customers faster, adapt to changing markets better, and grow more efficient



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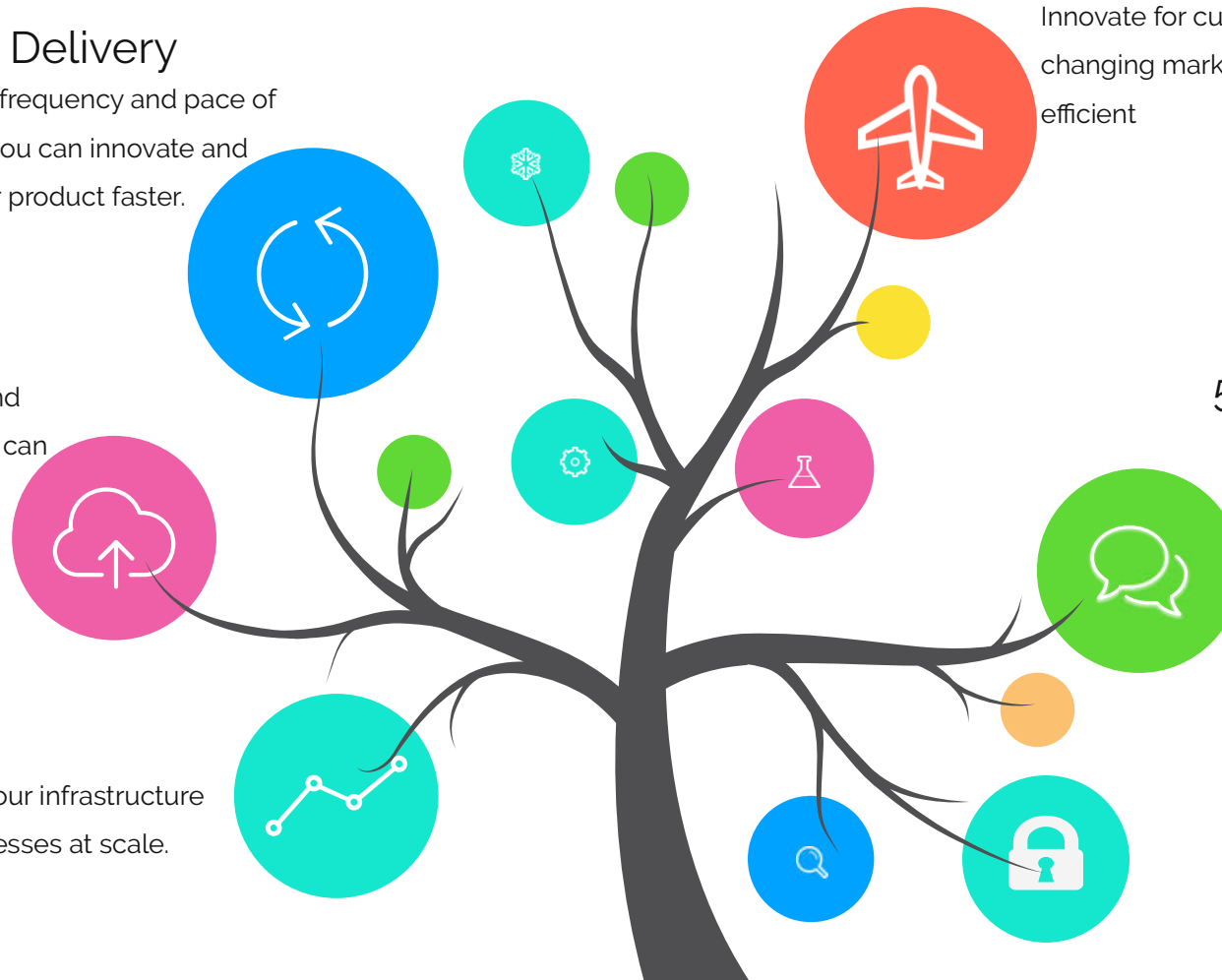
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Build more effective teams under a DevOps cultural model, which emphasizes values such as ownership and accountability.



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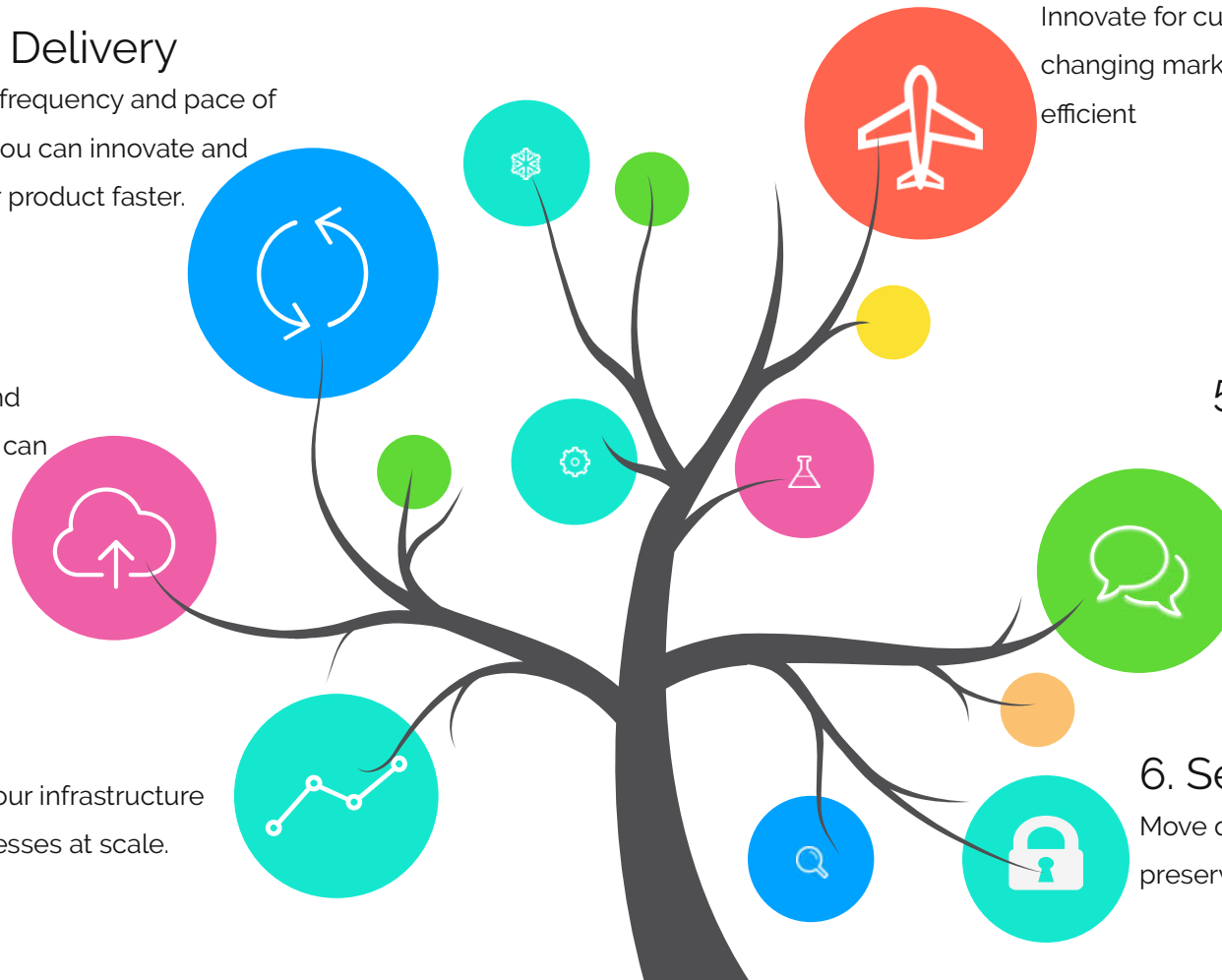
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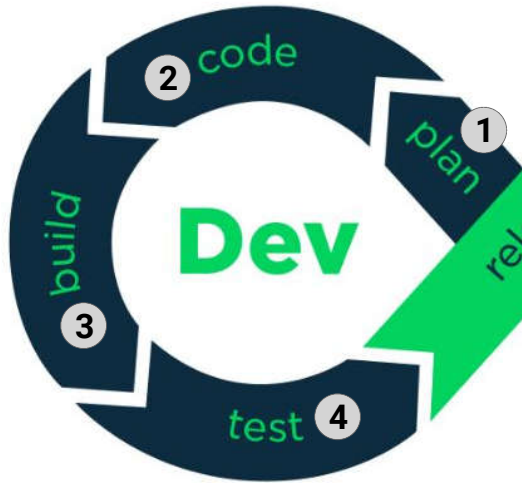
Build more effective teams under a DevOps cultural model, which emphasizes values such as ownership and accountability.

6. Security

Move quickly while retaining control and preserving compliance.



Understanding DevOps?



1. Dev

Development Phase.

Understanding DevOps?



2. Ops

Operations or Operating phase.

Understanding DevOps?

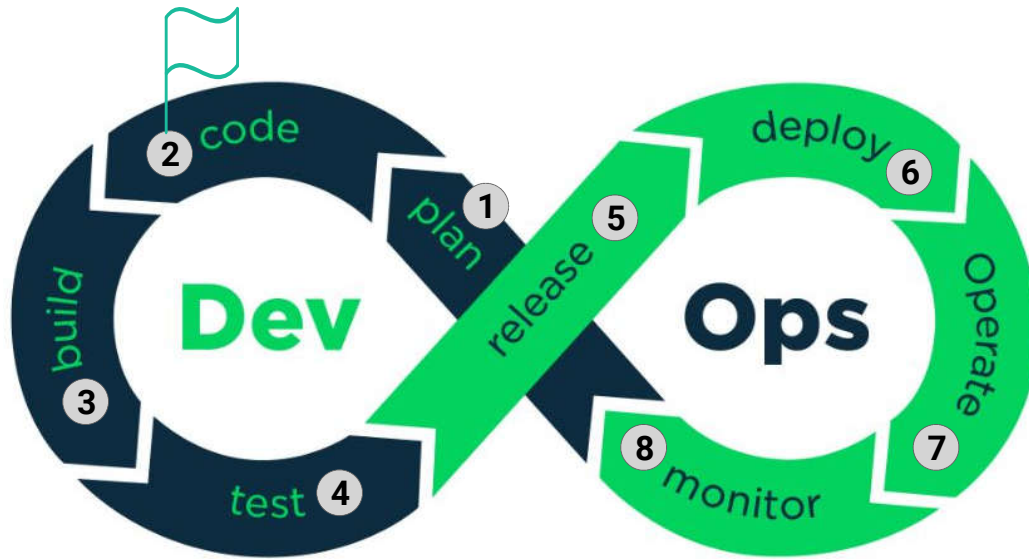


1. Plan

Planning phase involves a shorter goal planning. A scrum or agile planning is a better choice. The various tools you use are:

1. Microsoft Office, Google Docs / Sheet
2. Project Management Tools – Microsoft Project
3. Task Management tools – Asana, Jira, Mantis

Understanding DevOps?



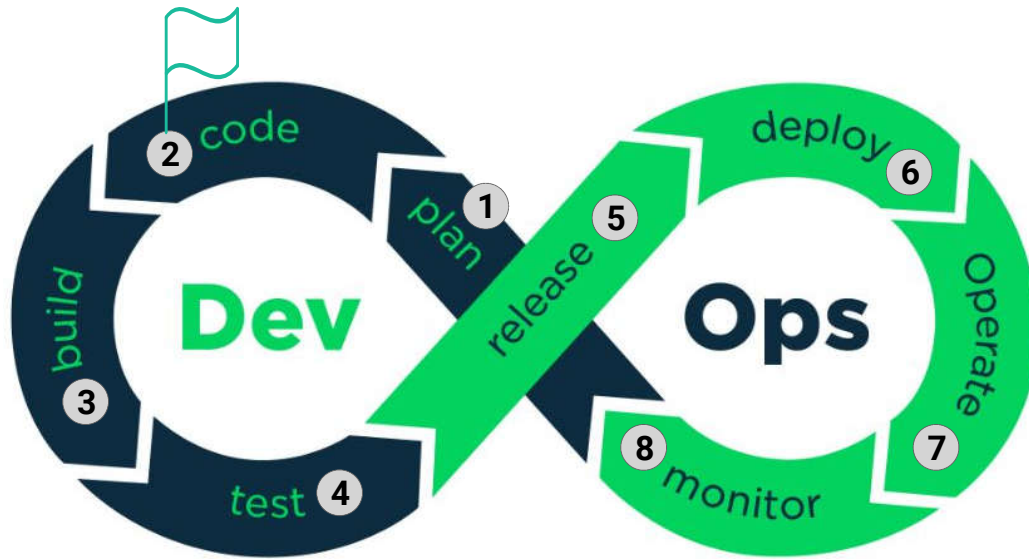
2. Code

1. Text Editors and Integrated Dev Environments:

- a. Sublime, Vim, Emacs ...
- b. Eclipse, XCode, Visual Studio...



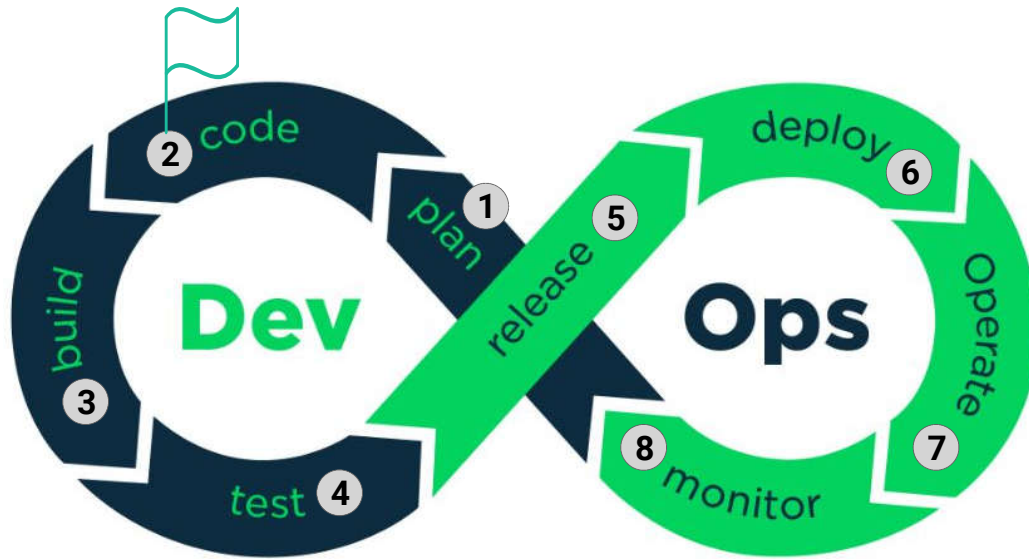
Understanding DevOps?



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 - a. Git – Best Choice, Older: SVN, CVS
 - b. Server – GitLab, GitHub or own.

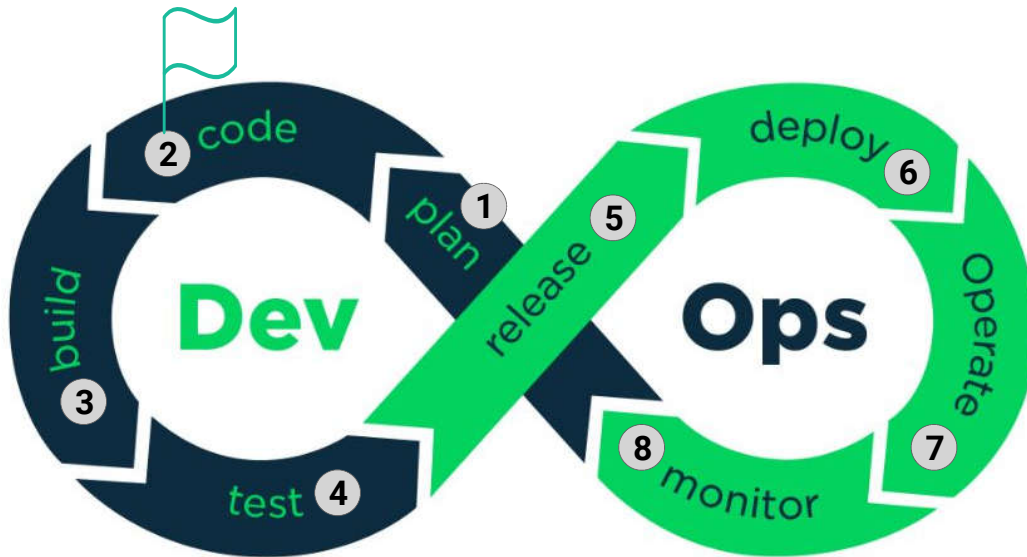
Understanding DevOps?



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 - b. Server – GitLab, GitHub or own.
3. Unit Test Case Libraries – depends on language
 - a. See [List of unit testing frameworks](#)
 - b. Used in "Test" phase too

Understanding DevOps?



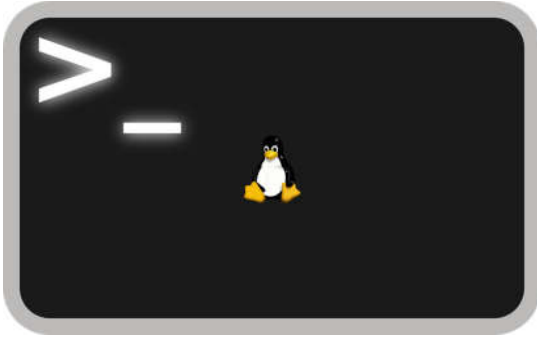
Coding - Hands On

We will do hands on with:

1. Linux Terminal
2. Git
3. Python



Coding – Hands On



Linux

1. Sign Up with CloudxLab.com
2. Go to "My Courses", Locate the [Course on "Linux"](#)
3. Or Go to "My Lab" and Open Jupyter
4. In Jupyter, open New -> Terminal
5. Here you can try all your commands. If you are new, go thru tutorial
6. Create a directory: ***mkdir devops***
7. Change the directory: `cd devops`

Git – Hands On



What is Git?

Git is a distributed version-control system for **tracking changes in source code** during software development. It is designed for **coordinating work among programmers**, but it can be used to track changes in any set of files. Its goals include speed, data integrity, and support for distributed, non-linear workflows.

Coding – Hands On



Try Git – Create a simple python program

1. Open Jupyter, go to the folder devops.
2. Create a text file with name "helloworld.py"
3. Add the following code to it:

```
print("Hello, World!")
```
4. Save the file

Coding – Hands On



Try Git

1. Go to the terminal and go to devops folder using "cd" command
2. Initialize repository: `git init` .
3. Add the python file create earlier: `git add *`
4. Commit: `git commit -m "This is hello world"`
5. See status: `git log`
6. Try changing the file and then reverting it with: `git checkout` .

Coding – Hands On



Try Git – Work with GitHub

1. Signup with GitHub – create login/pass and upload public key
2. Create a repository on github
3. Add origin
4. git push ...

Understanding DevOps?



3. Build

Build process involves compiling code, copying the assets, generating config and documentation. It should also work on developer machine as well as on unattended machine.

There are various build tools:

1. Maven, Ant, SBT etc..
2. Ref: ["Writing Spark Applications"](#)

Understanding DevOps?



4. Test

Testing involves verifying if the code is performance as per requirement. Unit Testing starts at coding time.

Write test cases before you code.

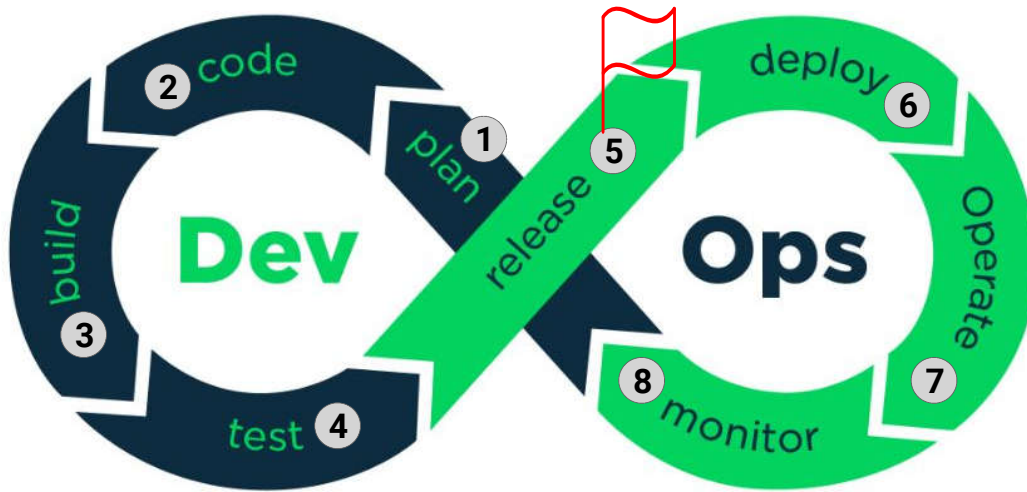
Various Types: Manual, Unit testing, Integration, Stress testing.

Tools: xUnit, Selenium, Scripts, [Stress testing](#)

See [List of unit testing frameworks](#)

To ensure completeness, we use [code coverage tools](#) like cobertura.

Understanding DevOps?



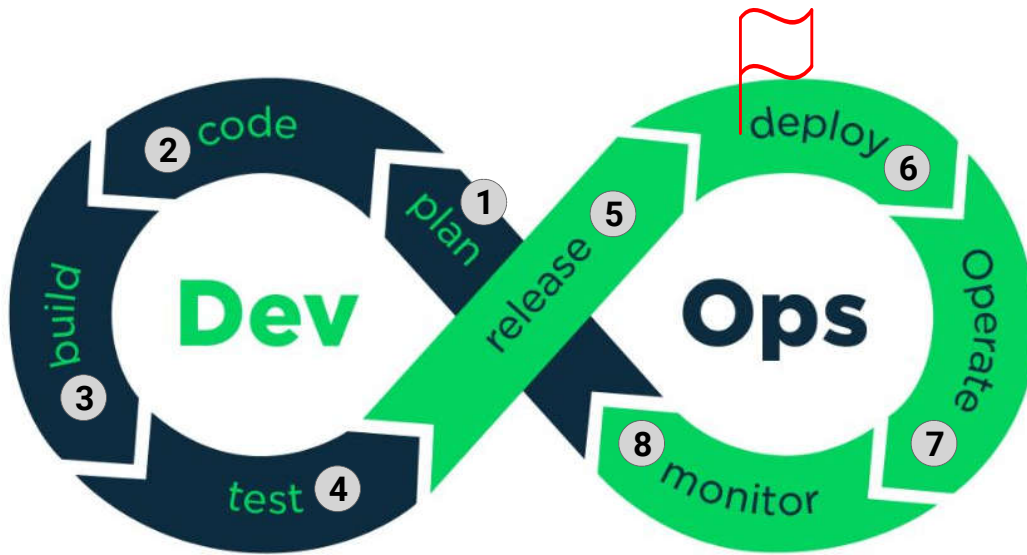
Jenkins

5. Release

Once the testing has been successfully done, the build is released as RC (release candidate) which is ready for being deployed in production.

Tools like Jenkins are used for releasing. Also, Apache Maven repositories are also used for releasing the binaries.

Understanding DevOps?



6. Deploy

Once the release is finalized, we can deploy it using different automation tools:

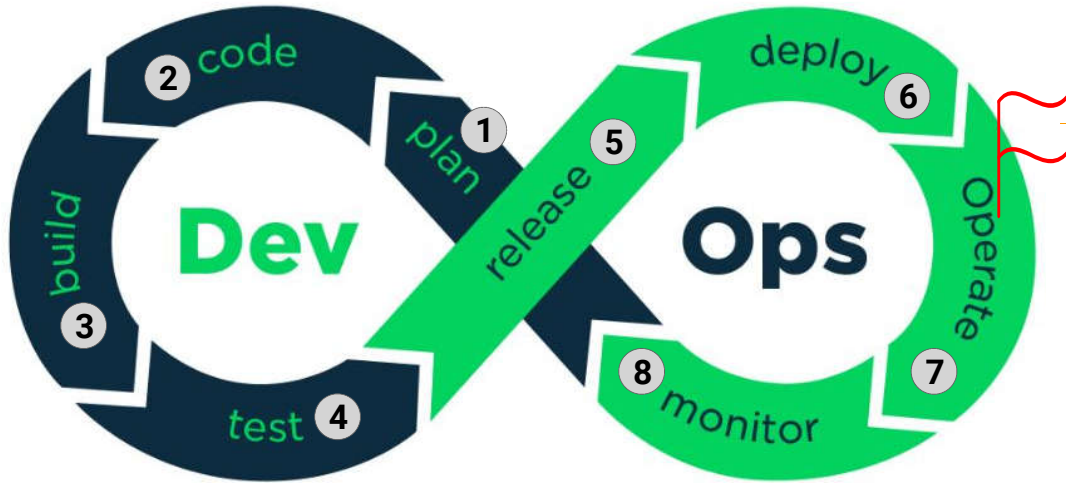
Puppet, Chef, Ansible, SaltStack

Newer tools such as Docker and Kubernetes help scale infinitely and instantaneously.

Docker and Kubernetes are used in testing too these days in CI.



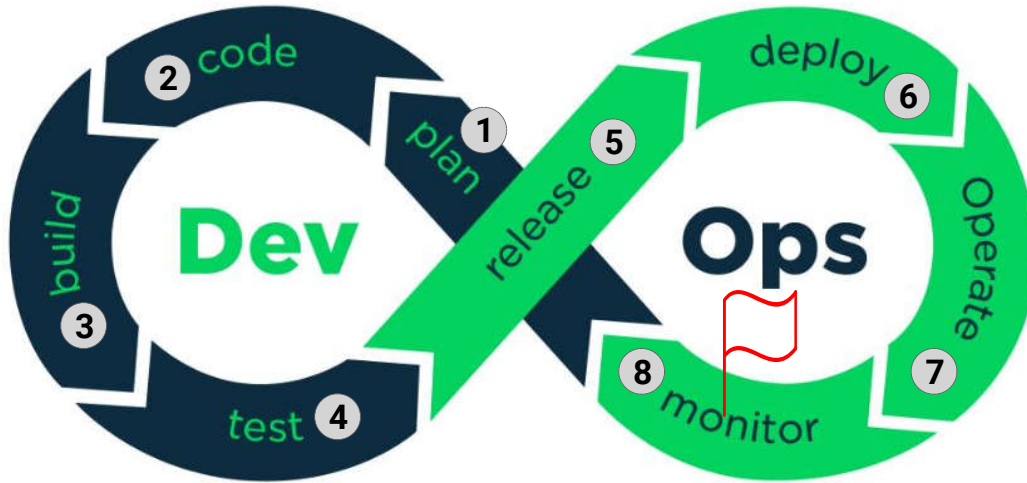
Understanding DevOps?



7. Operate

Once the software is in production, users can use it and the product managers can customize it. During the "operating" phase, we can measure the effectiveness using A/B Testing.

Understanding DevOps?



8. Monitor

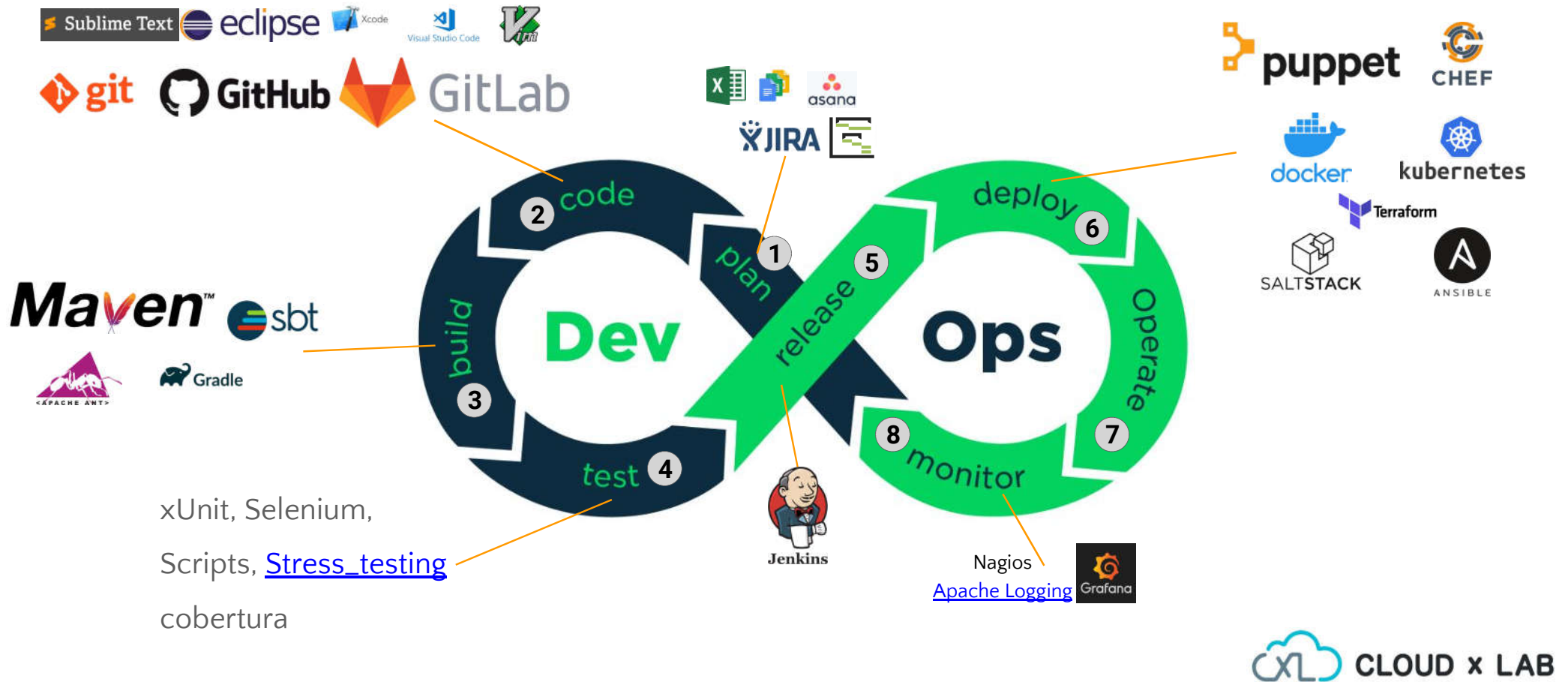
We also need to monitor the various system resource consumptions such as [Nagios](#).

We must also monitor the various logs and errors being thrown by the system.

See: [Apache Logging System](#)

Visualization tools such as Grafana are used to represent metrics.

Understanding DevOps?



DevOps Practices



Continuous Integration



Continuous Delivery



Microservices



Infrastructure as Code



Monitoring and Logging



Collaboration

Let's discuss the DevOps Best Practices

DevOps Practices



Continuous Integration

Continuous Integration



Continuous Delivery



Microservices



Infrastructure as Code



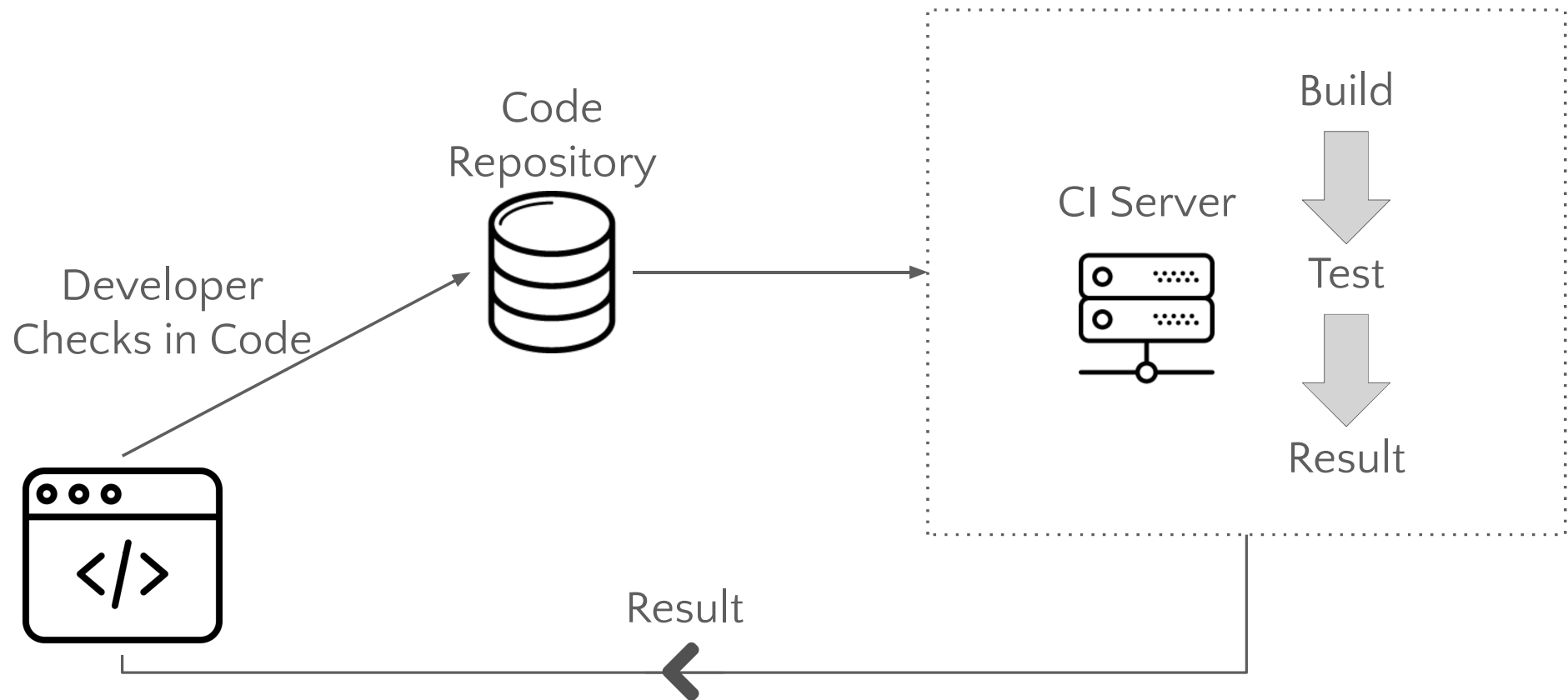
Monitoring and Logging



Collaboration

- Refers to the build and unit testing stages of the software release process
- Every code commit triggers an automatic workflow that
 - Builds the code and
 - Tests the code

DevOps Practices - Continuous Integration



DevOps Practices



Continuous Integration

Continuous Integration

- Helps in finding and addressing bugs quicker



Continuous Delivery



Microservices



Infrastructure as Code



Monitoring and Logging



Collaboration

DevOps Practices



Continuous Integration



Continuous Delivery



Microservices



Infrastructure as Code



Monitoring and Logging



Collaboration

Continuous Delivery

- Continuous Delivery automates the entire software release process.
- Continuous Delivery extends the Continuous Integration

DevOps Practices



Continuous Integration



Continuous Delivery



Microservices



Infrastructure as Code



Monitoring and Logging



Collaboration

Continuous Delivery

- Every code-commit triggers an automatic workflow that
 - Builds the code (Continuous Integration)
 - Tests the code and (Continuous Integration)
 - Then deploys the code to staging and then to production

DevOps Practices – Continuous Deployment

